Touhidul Alam Seyam

Software Engineer | Full-Stack and AI Systems

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PROFESSIONAL SUMMARY

Software Engineer with experience building and deploying full-stack applications, backend services, and AI-enabled products. Works across TypeScript, React, Next.js, Node.js, Python, and PostgreSQL; also develops open-source ML tooling and reproducible data-science pipelines. Author or co-author of ten source-backed journal, conference, chapter, and preprint works.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Go, Zig

Application Development: React, Next.js, Node.js, Django, FastAPI, REST APIs, React Native, Tailwind CSS

Data and Infrastructure: PostgreSQL, MongoDB, Redis, SQLite, Prisma, Convex, Docker, Git, AWS, Vercel

AI and Data: scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, computer vision, model evaluation, feature engineering

PROFESSIONAL EXPERIENCE

Software Engineer | Hello World Communications Ltd | Chattogram, Bangladesh | Aug 2024–Present

- Build, review, deploy, and maintain full-stack web applications for client and internal use across frontend, backend, and database layers.
- Collaborate with cross-functional stakeholders on requirements, implementation, production releases, and application support.
- Apply React, Next.js, Node.js, Python, PostgreSQL, and AI integrations to scalable application-development work.

Freelance Developer | Self-employed | Remote | Mar 2021–2025

- Deliver web applications from requirements and interface design through backend implementation, deployment, and ongoing support.
- Built digital products for small businesses, nonprofit and academic organizations, and event teams using modern JavaScript and Python stacks.

Research Assistant | BGC Trust University Bangladesh | Chattogram, Bangladesh | Jul 2023–2025

- Support faculty-led machine-learning research through experiment design, data analysis, implementation, academic writing, and publication preparation.
- Contributed to research spanning agricultural computer vision, healthcare risk modeling, malware classification, LLM performance, and high-performance clustering.

SELECTED PROJECTS

bun-scikit | TypeScript, Bun, Zig | github.com/Seyamalam/bun-scikit

- Built a scikit-learn-inspired ML library with native Zig acceleration, CI and benchmark gates, 209 tracked runtime exports, and documented model-selection, preprocessing, ensemble, clustering, and metrics APIs.

ASRRO Portal | Next.js, TypeScript, Convex, Better Auth | github.com/Seyamalam/asrro

- Developed a public website and role-aware organization-management system covering membership approvals, events, attendance, content, reporting, notifications, and restricted finance workflows.

Zodic | Python, PyPI | pypi.org/project/zodic/

- Published an MIT-licensed, zero-dependency Python validation package with chainable schemas, typed parsing, nested error reporting, transformations, unions, dates, enums, and framework-agnostic integration.

AgriScan | Next.js, TensorFlow, Computer Vision | doi.org/10.1186/s43067-024-00169-7

- Co-developed a cross-platform plant-disease diagnosis and crop-health system documented in a 2024 SpringerOpen journal article.

EDUCATION

B.Sc. (Hons.) in Computer Science and Engineering | BGC Trust University Bangladesh | Chattogram, Bangladesh | Expected Dec 2026

Research focus: software development, machine learning, computer vision, and applied AI.

SELECTED PUBLICATIONS

- *“AgriScan: Next.js powered cross-platform solution for automated plant disease diagnosis and crop health management.”* Journal of Electrical Systems and Information Technology, 2024. DOI: 10.1186/s43067-024-00169-7
- *“Comparing pre-trained models for efficient leaf disease detection: a study on custom CNN.”* Journal of Electrical Systems and Information Technology, 2024. DOI: 10.1186/s43067-024-00137-1
- *“Next-Generation K-Means Clustering: Mojo-Driven Performance for Big Data.”* International Journal of Intelligent Information Systems, 2025. DOI: 10.11648/j.ijiis.20251401.12
- *“Architectures for AI-Driven Visual Assistance: Evaluating Server-Mediated Mobile and Direct Access Desktop Client Implementations.”* Lecture Notes in Networks and Systems, 2026. DOI: 10.1007/978-3-032-15764-5_45

SELECTED CERTIFICATIONS

IBM Applied Data Science with Python; IBM Deep Learning with TensorFlow; IEEE Computer Society Machine Learning Mastery; Cisco Python Essentials 1 and 2; Harvard CS50 for Educators